Thermal History Shapes Coral Bleaching Responses

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Abstract. Marine heatwaves in 2016 and 2017 caused mass coral bleaching across the Great Barrier Reef.

Reefs that had experienced moderate heat stress in 2016 showed 31% lower bleaching mortality in 2017.

This pattern suggests thermal acclimation can partly buffer corals against repeated heatwaves.

Discussion. Acclimation benefits disappeared when heat stress exceeded eight degree heating weeks. Ocean warming projections imply that severe bleaching events will occur annually on most reefs by 2050, outpacing acclimation.